

Igor Vilghelm

Full stack developer

CONTACT

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Vilnius, Lithuania

EDUCATION

Vilnius University
Master's degree in Computer Modeling
2019 – 2021

Vilnius University
Additional studies
2018 – 2019

Vilniaus Kolegija
Bachelor's degree, Information systems engineer
2014 – 2017

LANGUAGE

Lithuanian / English / Russian

ABOUT ME

Senior Full-Stack Developer with 10+ years of experience building large-scale web systems for energy metering and data processing.

Experienced in system architecture, data visualization, cloud deployment, and CI/CD automation. Led development of complex SCADA-like interfaces and backend systems working with high-volume relational databases and complex data models.

Strong focus on performance, clean architecture, and scalable systems.

EXPERIENCE

Full stack developer / Team lead 2016-Present
UAB Sigma Telas

Team Lead & Full Stack Developer (2024 – Present)

- Assign tasks to team members and manage workload distribution
- Review code and deliverables to ensure quality and consistency
- Support team members in solving technical challenges and blockers
- Coordinate with cross-functional teams (e.g., DevOps, DB) when needed
- Monitor progress and ensure timely delivery of features and tasks
- Participate in technical decision-making and architecture discussions
- Conducted post-audit security assessments and vulnerability scans using OWASP ZAP, identifying and remediating critical backend security flaws to ensure regulatory compliance
- Improve team processes and development workflows
- Actively contribute to the codebase (backend & frontend development)

Full Stack Developer (2016 – 2024)

- Worked in multiple projects across different domains and business areas
- Designed and developed applications from scratch, including architecture and core systems
- Delivered end-to-end features from concept to production
- Improved application performance and reliability through optimizations, bug fixing, and refactoring
- Implemented features based on business and client requirements
- Maintained and improved existing codebase
- Built and integrated backend services and APIs
- Developed responsive and user-friendly frontend interfaces
- Set up and maintained CI/CD pipelines to streamline development and deployment
- Ensured code quality through testing, code reviews, and best practices

SKILLS

- Software development
  - Front-end development
  - Back-end development
- Automation (CI / CD)
- Cloud computing
  - Cloud management
- Management
- Design

STACK

Core stack
- JavaScript / TypeScript
- React / React Native / Redux
- Node.js
- PHP (modern + legacy)

Backend
- Phalcon 5 (PHP framework)
- REST APIs
- High-load systems
- Data processing

Databases
- Oracle / MariaDB / PostrgeSQL
- Query optimization
- Stored procedures

DevOps
- Jenkins CI/CD
- IIS / Nginx / Apache

Additional
- Python 3
- Git

PROJECTS

AI Assistant Backend Service for Web Application (2026 – Present)
- Developed a backend service using Node.js (Fastify) to power an AI assistant for the Web Energy Management System, enabling users to analyze system events and receive actionable recommendations via a dedicated 'Ask AI' interface.
- Integrated LLM-based solutions (Ollama / LM Studio) to provide contextual explanations of system behavior and assist in decision-making.
- Designed scalable APIs and integrated AI capabilities into the existing system architecture.

Mobile Energy Management System (2025 – Present)
- Designed and developed a mobile energy management application using React Native (Expo) for field engineers, enabling real-time monitoring and data access on the go.
- Focused on performance optimization, offline-first capabilities, and seamless integration with existing backend services.

Web Energy Management System – Maintenance & Evolution (2020 – Present)
- Maintained and evolved a high-load energy management system, ensuring scalability, performance, and reliability in production environments.
- Worked with high-volume relational databases and complex data models.

Web Energy Management System – Architecture & Development (2017 – Present)
- Led the design and development of a smart metering system for automated commercial energy accounting.
- Designed and implemented system architecture (frontend & backend), including reusable component libraries and scalable module structure.
- Built a custom component system using React and Redux, enabling consistent UI development across the application.
- Implemented robust Content Security Policy (CSP) rules and hardened IIS security headers to secure frontend assets against client-side injection and cross-site attacks.
- Integrated Redis / Garnet for login rate-limiting and server-side session invalidation, safeguarding databases from brute-force lockouts and enforcing global token revocation.
- Conducted security testing and vulnerability remediation with OWASP ZAP post-cybersecurity audit, resolving critical backend security flaws.
- Developed and maintained backend endpoints for communication with Oracle and MariaDB databases and data acquisition systems (REST APIs).
- Implemented real-time data visualization, including SCADA-like SVG-based interactive scheme viewer & editor with live data binding.
- Worked with large-scale data processing, including Big Data and IoT integrations, data analysis, and data visualization pipelines.
- Optimized application performance and query execution, including analysis and tuning of complex database queries, procedures, and functions.
- Integrated and adapted third-party npm components (charts, grids) to meet specific business requirements.
- Set up and maintained CI/CD pipelines using Jenkins, including custom automation workflows.
- Contributed to cloud adaptation, SSO integration, and infrastructure improvements.
- Provided technical guidance to team members and participated in architecture decisions.

Legacy Web Application Support & Migration (2016 – 2018)
- Maintained and supported a legacy web application, ensuring system stability and resolving production issues.
- Worked on bug fixing, performance improvements, and gradual refactoring of legacy codebase.

KEY ACHIEVEMENTS

- Delivered and deployed energy management systems across multiple regions, including Lithuania and Latvia at different operational levels, as well as national-scale implementations in Central Asia.
- Built and scaled a high-load platform processing data from 1M+ metering devices and millions of data points.
- Engineered a time-series data system supporting per-second to monthly aggregation levels.
- Delivered high-performance data visualization (grids, charts, SCADA-like UI), ensuring smooth UX with large-scale datasets.
- Led the architecture and development of a SCADA-like real-time system for energy data visualization and processing.
- Built and scaled a high-load data processing system handling large volumes of metering and IoT data.
- Designed and implemented a reusable frontend architecture (React + Redux), improving development consistency and scalability.
- Optimized database queries and backend performance for complex Oracle and MariaDB systems.
- Established CI/CD pipelines using Jenkins, improving development and deployment workflows.
- Developed a mobile application (React Native) with offline-first capabilities for field engineers.
- Designed and implemented a backend service for AI-powered data querying within the energy management system.
- Implemented and integrated an AI Assistant interface into the enterprise web frontend, providing automated contextual data summaries, event analysis, and operational recommendations.
- Engineered ML-based anomaly detection and time-series forecasting for metering archives, paired with automated LLM summaries of detected anomalies.
- Implemented distributed rate-limiting and centralized session management using Redis / Garnet, protecting authentication endpoints from brute-force/DDoS attacks and enabling immediate multi-device token revocation.